PV101 Day 7 Student Worksheet Answer Key

Instructor copy

Day_07_PV101_Load_Assessment_and_System_Sizing_2026-07-15

RV training loads

Load	Calculation	Wh/day
3 interior LED lights	3 x 8 W x 4 h	96
Water pump	60 W x 0.5 h	30
Vent fan	36 W x 3 h	108
Phone/USB charging	24 W x 2 h	48
Laptop charger (AC)	65 W x 2 h	130
Total	96 + 30 + 108 + 48 + 130	412 Wh/day

Storage

- Required stored energy: 412 Wh/day x 1 day = 412 Wh.
- Battery nameplate energy: 412 Wh / 0.80 = 515 Wh.
- Basic capacity at 12 V: 515 Wh / 12 V = 42.9 Ah.
- Accept clear rounding. Students must state the 1-day autonomy and 80% usable-fraction / maximum-depth-of-discharge assumptions.

Array and inverter

- Raw array watts: 412 Wh / 4 peak sun hours = 103 W.
- Adjusted exercise result: 103 W x 1.25 = 128.75 W, or about 129 W.
- Concurrent AC load: 65 W for the laptop charger; the other listed loads are treated as DC in this exercise.
- Minimum inverter exercise result: 65 W x 1.25 = 81.25 W.
- Do not turn the result into a product recommendation. Actual selection needs real specifications, surge data, and documented design inputs.

One-line diagram acceptance

Accept a clearly labeled path such as PV -> charge controller -> battery -> protected DC load/inverter branches. Require complete return paths, source-side protection, disconnects, polarity/labels, and logical meter points. Conceptual variants are acceptable when electrically coherent and safely explained.